SUBJECT	Lower Secondary Maths Year 8
TOPIC	Module Nine
ASSIGNMENT	Nine
STUDENT NAME	

FEEDBACK AND REFLECTION:

- ☐ I have read the feedback on my last assignment carefully.
- ☐ I have considered whether any of that feedback is relevant to this assignment.

One thing I will try to do either the same or differently in this assignment as a result of my previous feedback is: ____

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ASSIGNMENT GUIDANCE

- Make sure you write your full name clearly in the box at the top of this page.
- Please **write** your answers **by hand** in **black ink** on this question paper, unless you have obtained specific permission to submit typed work.
- This is not a timed assignment. We estimate that this assignment may take you between 30 and 40 minutes. However, if you need to take longer, then please do so.
- You are not expected to complete your assignments without referring to your notes or coursebook.
- In order for maximum marks to be awarded, for any question worth more than one mark, please show your workings to make your method clear.
- Calculators are **not** to be used in this assignment apart from in the last question. You will need a pencil and ruler.

LS Maths Assignment Nine

1. Show how you would use the clever percentages trick, taught in this module, to find the following. (Be sure to make your method clear and also work out the actual answer).

   a. What is 4% of 50? (2 marks)

   b. What is 80% of 25? (2 marks)

   c. What is 230% of 10? (2 marks)

(Total 6 marks)

2. Jenna and Sally help their father sell some things at a car boot sale, which raises £88.
   Their father says he will give them 25% of the takings each.

   Sally works out 25% of £88
   Jenna says it will be easier to work out 88% of £25 instead.

   Show how both girls could use a non-calculator method to find the amount they will each receive. Which sister used the easiest method?

(Total 6 marks)

3. Join each decimal with a straight line to its fraction equivalent.

0.3333333……	$\frac{323}{999}$
0.323232323…..	$\frac{3}{10}$
0.3	$\frac{8}{25}$
0.323323323…	$\frac{32}{99}$
0.32	$\frac{1}{3}$
0.232323…	$\frac{23}{99}$

(6 marks)

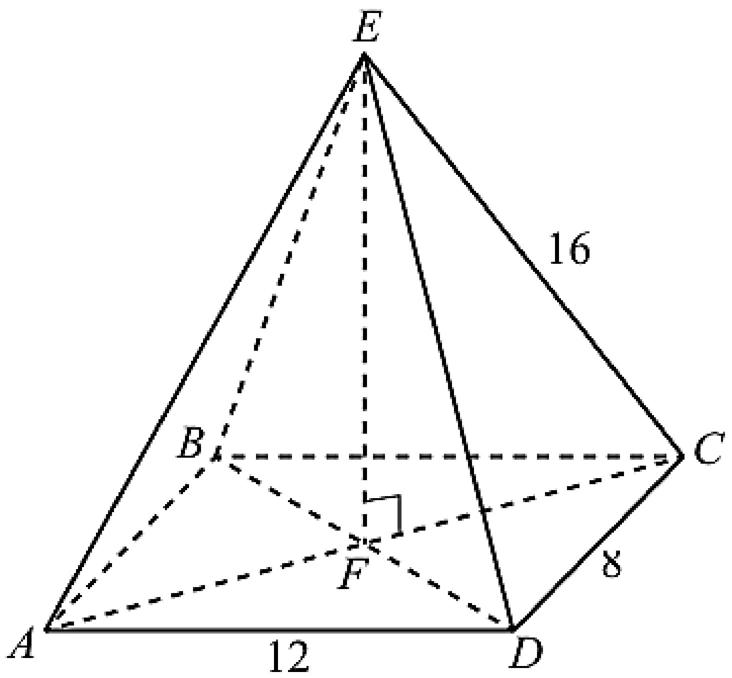
4. The number of issues of books from a library are given for the years 2001–2011. The numbers are in hundreds of thousands.

2001	**2002**	**2003**	**2004**	**2005**	**2006**	**2007**	**2008**	**2009**	**2010**	**2011**
125	131	122	118	125	129	132	135	128	123	120

Plot a time series graph of this data. **Hint**: Choose carefully how to restrict the range of values on your axes so that your plotted graph fills much of this paper.

(5 marks)

5. ABCDE is a **rectangular-based** pyramid.



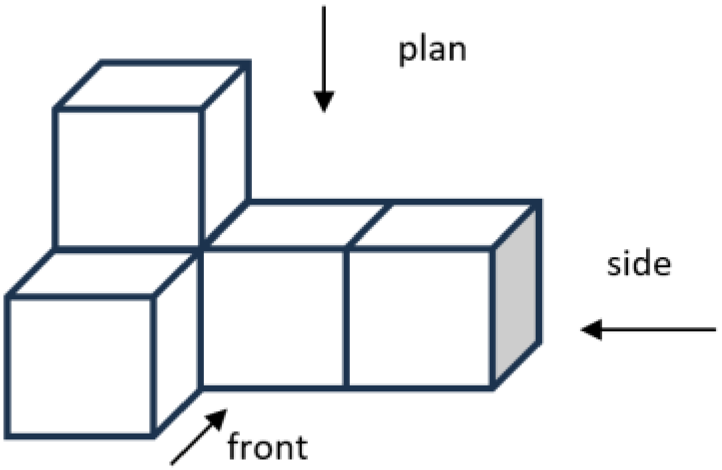
a. How many planes of symmetry does this pyramid have? (1 mark)

b. Another pyramid is built with a **square** base. How many planes of symmetry will this pyramid have? Why is this a different to the number of planes of symmetry of the rectangular-based pyramid?

(2 marks)

(Total 3 marks)

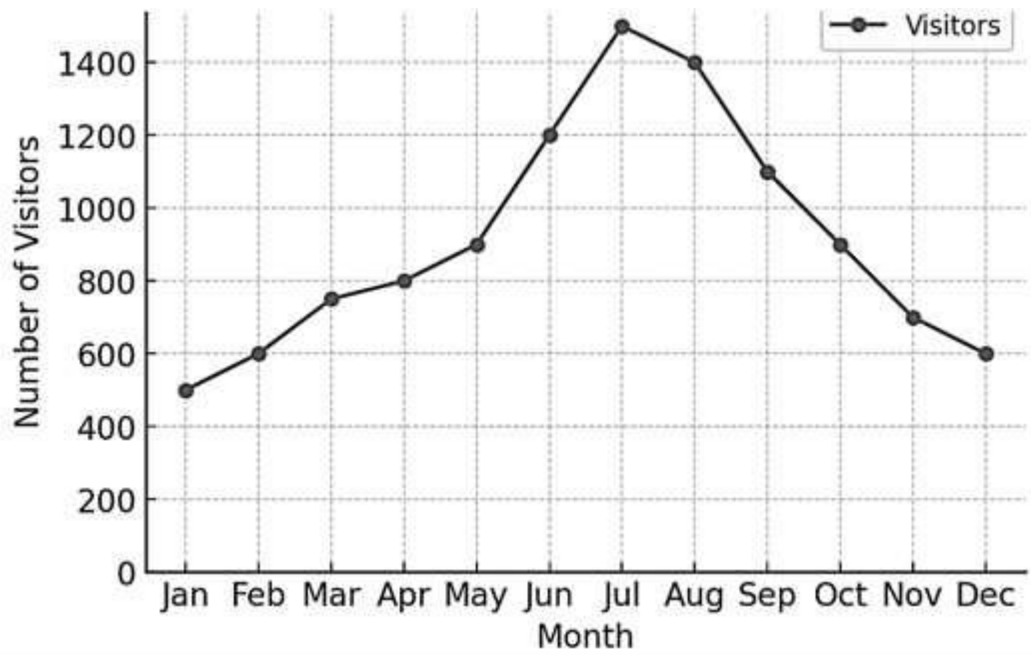
6. Draw the plan, front and elevation of the solid below.



Plan Front Side

(3 marks)

7. The graph below shows the number of visitors to a park over a year.



a. In which month were the most visitors recorded? (1 mark)

b. Describe the overall trend in visitor numbers throughout the year. (1 mark)

c. Identify any months where visitor numbers dropped significantly and suggest possible reasons. (2 marks)

d. If this pattern continues, estimate the number of visitors in January of the next year. (1 mark)

(Total 5 marks)

8. This stem and leaf diagram shows the ages of people on a bus.

0	2 3 4 5
1	3 3 5 8 8
2	0 3 4 4 4 5 5 5 6 7 9
3	1 3 7
4	2 8 9
5	
6	0
7	0 6 7

Key 2|5 represents 25 years old.

a. Find the median age, making your method clear. (2 marks)

b. Find the range of ages and the mode(s). (2 marks)

c. Find the mean age of all the people on the bus, showing your method. You may use a calculator to help with the calculations. (2 marks)

(Total 6 marks)

TOTAL FOR ASSIGNMENT 40 MARKS